CONNECTED FIBRES GIVE A BIJECTION BETWEEN THE CONNECTED COMPONENTS OF SOURCE AND TARGET

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that a connected component of a space is closed and that a continuous image of a connected space is connected. Invoked inside the counted proof (source lines 950-954). Justifies taking preimages of components and comparing connected subsets.
- The hypothesis that preimages of points under the map are connected. Used inside the counted proof (source lines 955-958). Splits the preimage of a component along the fibres.

Source reference. Original lemma lemma-connected-components: sources/stacks-topology/pinned-original-source.tex, source lines 875–875. Navigation pointer only: the counted text refers to the Stacks item lemma-connected-components, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-image-connected-space: sources/stacks-topology/pinned-original-source.tex, source lines 856–856. Navigation pointer only: the counted text refers to the Stacks item lemma-image-connected-space, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces. Assume that*

(1) *all fibres of f are connected, and*
(2) *a set $T \subset Y$ is closed if and only if $f^{-1}(T)$ is closed.*

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Then f induces a bijection between the sets of connected components of X and Y.

Proof. Let $T \subset Y$ be a connected component. Note that T is closed, see Lemma lemma-connected-components. The lemma follows if we show that $f^{-1}(T)$ is connected because any connected subset of X maps into a connected component of Y by Lemma lemma-image-connected-space. Suppose that $f^{-1}(T) = Z_1 \amalg Z_2$ with Z_1, Z_2 closed. For any $t \in T$ we see that $f^{-1}(\{t\}) = Z_1 \cap f^{-1}(\{t\}) \amalg Z_2 \cap f^{-1}(\{t\})$. By (1) we see $f^{-1}(\{t\})$ is connected we conclude that either $f^{-1}(\{t\}) \subset Z_1$ or $f^{-1}(\{t\}) \subset Z_2$. In other words $T = T_1 \amalg T_2$ with $f^{-1}(T_i) = Z_i$. By (2) we conclude that T_i is closed in Y. Hence either $T_1 = \emptyset$ or $T_2 = \emptyset$ as desired. $\square$